9.1 Lines Cutting the Plane

WORKSHEET

The question

STEP II 1990, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q4

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Pause and try Draw the cases $n = 1, 2, 3, 4$ carefully, keeping the lines in general position (no two parallel, no three through a point). Count the segments and the regions in each case. Can you guess the patterns?

STAGE 1 Counting segments

Ask yourself Concentrate on one line. How many of the other lines cross it, and into how many pieces do they cut it?

STAGE 2 What happens when we add a line?

Ask yourself Suppose $n - 1$ lines are drawn, and you add the nth. It is cut into n segments by the others. What does each segment do to the region it passes through?

What's really going on

Euler's formula in disguise

Here is a second proof, which explains the answer rather than just deriving it. Think of the arrangement of lines as a map: the crossing points are *vertices*, the segments are *edges* and the regions are *faces*. There are $V = \binom{n}{2}$ vertices (one for each pair of lines) and $E = n^2$ edges. The infinite segments have only one end, so add a single extra vertex "at infinity" where they all meet; now the picture is a map drawn on a sphere (imagine the plane wrapped round a ball, with infinity at the top). Euler's formula $V - E + F = 2$ from Section 7.9 gives

$$\left(\binom{n}{2} + 1\right) - n^2 + F = 2, \qquad \text{so} \qquad F = n^2 - \frac{n(n-1)}{2} + 1 = \frac{n^2 + n + 2}{2}.$$

The same number, from a completely different direction. The regions also split neatly into types: $2n$ of them are unbounded (one between each pair of neighbouring infinite rays), and the remaining $\frac{1}{2}(n^2 + n + 2) - 2n = \binom{n-1}{2}$ are bounded.

Binomial coefficients and higher dimensions

The formula can be written as

$$R(n) = \binom{n}{0} + \binom{n}{1} + \binom{n}{2},$$

and this is not a coincidence. Turn the picture so that no line is horizontal. A region that is bounded below has a lowest point, which must be a crossing, and each crossing is the lowest point of exactly one region (the one directly above it): that gives $\binom{n}{2}$ regions, one for each pair of lines. Every other region stretches down for ever, and so meets a horizontal line drawn far below all the crossings; the n lines cut that horizontal line into $n + 1$ intervals, one in each such region. In total, $\binom{n}{2} + n + 1$. In three dimensions, n planes in general position cut space into

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} = \frac{n^3 + 5n + 6}{6}$$

pieces (the *cake numbers* $1, 2, 4, 8, 15, 26, \ldots$), and the pattern continues in every dimension. The proof is the same induction: a new plane is cut by the others into regions arranged like lines in a plane, and each of those regions splits a piece of space in two.

Steiner and the arrangement of hyperplanes

The formula for regions cut by lines and planes was found by Jakob Steiner in 1826. Today it is the starting point of the theory of *hyperplane arrangements*, which counts the regions formed by arbitrary collections of lines, planes and their higher-dimensional analogues, including non-general ones. A remarkable 1975 theorem of Thomas Zaslavsky counts the regions of any arrangement using a polynomial that encodes

which hyperplanes meet. The same ideas appear in machine learning: a neural network built from "ReLU" units cuts its input space into regions by hyperplanes, and the number of regions measures how complicated a function it can represent.

Try this next

1. **Circles.** Show that n circles, each pair meeting in two points and no three through a point, divide the plane into $n^2 - n + 2$ regions.

2. **Parallel lines.** If the n lines include k that are all parallel to each other (and otherwise general position), how many regions are there?

3. **Cake numbers.** Prove by induction that n planes in general position divide space into $\frac{1}{6}(n^3 + 5n + 6)$ regions.

4. **A pattern that breaks.** Put n points on a circle and join every pair with a chord, with no three chords through one point inside the circle. For $n = 1, 2, 3, 4, 5$ you get $1, 2, 4, 8, 16$ regions. Count the regions for $n = 6$. (*Hint:* use Euler's formula; each set of four points gives one crossing.) Show that the number is $1 + \binom{n}{2} + \binom{n}{4}$.

Further reading

- **Ronald Graham, Donald Knuth & Oren Patashnik, *Concrete Mathematics*** (Addison-Wesley), Chapter 1. Starts with exactly this problem, and the "pizza" and "zig" variants.
- **Miklós Bóna, *A Walk Through Combinatorics*** (World Scientific). A friendly first course, with chapters on induction, graphs and Euler's formula.
- **Richard Guy, "The strong law of small numbers"**, *American Mathematical Monthly* 95 (1988). Many patterns that look obvious and turn out to be false, including the circle problem in the exercises.

The full guided solution is in Section 9.1:
https://beyondstep.keithmcnulty.org/ch09/lazy-caterer.html